



LUND UNIVERSITY

Characterizing the effects of extreme heat events on all-cause mortality a case study in Ahmedabad city of India, 2002–2018

Sharma, Ayushi; Dutta, Priya; Shah, Priyanka; Iyer, Veena; He, Hao; Sapkota, Amir; Gao, Chuansi; Wang, Yu Chun

Published in:
Urban Climate

DOI:
[10.1016/j.uclim.2024.101832](https://doi.org/10.1016/j.uclim.2024.101832)

2024

Document Version:
Peer reviewed version (aka post-print)

[Link to publication](#)

Citation for published version (APA):

Sharma, A., Dutta, P., Shah, P., Iyer, V., He, H., Sapkota, A., Gao, C., & Wang, Y. C. (2024). Characterizing the effects of extreme heat events on all-cause mortality: a case study in Ahmedabad city of India, 2002–2018. *Urban Climate*, 54, Article 101832. <https://doi.org/10.1016/j.uclim.2024.101832>

Total number of authors:
8

Creative Commons License:
CC BY-NC-ND

General rights

Unless other specific re-use rights are stated the following general rights apply:
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the public portal

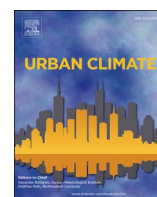
Read more about Creative commons licenses: <https://creativecommons.org/licenses/>

Take down policy

If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

LUND UNIVERSITY

PO Box 117
221 00 Lund
+46 46-222 00 00



Characterizing the effects of extreme heat events on all-cause mortality: A case study in Ahmedabad city of India, 2002–2018

Ayushi Sharma^{a,b}, Priya Dutta^c, Priyanka Shah^c, Veena Iyer^c, Hao He^d,
Amir Sapkota^e, Chuansi Gao^f, Yu-Chun Wang^{b,g,*}

^a Department of Civil Engineering, College of Engineering, Chung Yuan Christian University, 200 Chung-Pei Road, Zhongli 320, Taiwan

^b Department of Environmental Engineering, College of Engineering, Chung Yuan Christian University, 200 Chung-Pei Road, Zhongli 320, Taiwan

^c Indian Institute of Public Health Gandhinagar (IIPHG), Public Health Foundation of India (PHFI), Near Lekwada Bus Stop, Gandhinagar 382042, Gujarat

^d Dept. of Atmospheric and Oceanic Science, University of Maryland, College Park, MD, United States

^e Departement of Epidemiology and Biostatistics, University of Maryland, School of Public Health, College Park, MD 20742, United States

^f Thermal Environment Laboratory, Ergonomics and Aerosol Technology, Department of Design Sciences, Faculty of Engineering (LTH), Lund University, Sweden

^g Research Center for Environmental Changes, Academia Sinica, 128 Academia Road, Section 2, Nankang, Taipei 11529, Taiwan

ARTICLE INFO

Keywords:

Extreme heat events (EHEs)

Mortality

Exposure-response

Minimum Mortality Temperature (MMT)

DLNM

Ahmedabad

ABSTRACT

Recent years have seen a rise in extreme heat event (EHE)-related deaths in India. However, the impact of specific temperature thresholds on health risks remains understudied. Using Distributed Lag Non-Linear Models (DLNM), we explored the link between EHEs, defined by various temperature thresholds, and mortality risk in Ahmedabad, India, from 2002 to 2018, considering a 21-day lag. We observed a 'J'-shaped exposure-response curve, identifying a Minimum Mortality Temperature (MMT) of 26 °C for Ahmedabad. Notably, a higher and sustained risk of all-cause mortality was associated with $T_{\max} > 35$ °C. EHEs definition of $T_{\max} \sim 40$ °C (95th percentile) increased all-cause mortality risk by 30% (Relative Risk (RR): 1.30, 95% Confidence Interval (95% CI): 1.26–1.35), with substantially higher risk at $T_{\max}$ of 45 °C (RR: 3.08, 95% CI: 2.47–3.83). Analysis of attributable fractions (AF) indicated $T_{\max} \geq 85$ th percentile contributed most to total mortalities, with an AF of 3.58% (95% CI: 3.20–3.96). Gender-stratified analysis revealed higher risk of EHE-related deaths for females. The highest mortality risk was identified on the same day of exposure and persisted longer during more intense EHEs. The activation of city's heat action plans should consider the significantly elevated mortality risk below the current threshold (~ 40 °C) and the persistent risk during high-intensity EHEs.

1. Introduction

The most recent Intergovernmental Panel on Climate Change (IPCC) report concluded that extreme heat events (EHEs) have become more frequent, intense, and long-lasting across the globe (*Sixth Assessment Report of Intergovernmental Panel on Climate Change (IPCC) Climate Change 2021, The Physical Science Basis, 2021*). Numerous studies worldwide have explored the temperature-mortality

* Corresponding author at: Department of Environmental Engineering, College of Engineering, Chung Yuan Christian University, 200 Chung-Pei Road, Zhongli 320, Taiwan.

E-mail address: ycwang@cycu.edu.tw (Y.-C. Wang).

<https://doi.org/10.1016/j.uclim.2024.101832>

Received 16 November 2022; Received in revised form 6 November 2023; Accepted 4 February 2024

Available online 21 February 2024

2212-0955/© 2024 Elsevier B.V. All rights reserved.

relationship, consistently revealing a heightened mortality risk with extreme temperatures (Dimitrova et al., 2021; Hashizume et al., 2009; Parks et al., 2020; Gasparrini et al., 2015; Azhar, 2019; Dutta et al., 2020). Some have even reported significant excess heat-associated mortality during the heatwave period (Guo et al., 2018; Fouillet et al., 2006; Knowlton et al., 2009). For instance, the 2003 heat waves in France resulted in 15,000 excess mortality (Fouillet et al., 2006). Similarly, the 2006 heatwaves in California caused around 16,166 excess emergency department visits and 1182 excess hospitalizations across the state (Knowlton et al., 2009). However, there remains a critical knowledge gap regarding temperature-associated health risks in low-and-middle-income countries (LMICs). These countries often contend with tropical to sub-tropical climates characterized by elevated temperature thresholds.

Cities and urban areas in LMICs are particularly susceptible to EHEs due to the exacerbating effect of Urban Heat Island (UHI) (Heaviside et al., 2017). In these regions, rapid urban development has transformed the open spaces into heat-absorbing concrete expanses, resulting in higher UHI effects (Azhar et al., 2014). Historical evidence emphasizes the heightened vulnerability of populations to EHEs in urban areas (Wang et al., 2019). Recent studies from India show that heat wave events increased all-cause mortality by up to 17% in the city of Hyderabad during 2006–2015 (Rathi and Sodani, 2021) and up to 40% in Nagpur during the 2010–2014 (Dutta et al., 2020), with limited evidence available from rural areas. Likewise, Ahmedabad city of India where South Asia's first heat action plan (HAP) was implemented has also experienced heat-related excess mortalities (Azhar et al., 2014; Hess et al., 2018; Wei et al., 2021). A major heatwave event in May 2010 resulted in 1344 additional deaths (Azhar et al., 2014).

However, the impact of EHEs varies across geographic regions and is influenced by several factors, like access to air conditioning, demographic compositions (Wang et al., 2019), and thermal acclimatization of the local population, often represented by the minimum mortality temperature (MMT) (López-Bueno et al., 2021). Yet, most of the studies in the region have consistently reported excess heat-related mortalities (Azhar, 2019; Azhar et al., 2014; Nori-Sarma et al., 2019), and there is a paucity of data on how the use of specific thresholds to define EHEs affects the observed association between heat waves and mortality in India. The lack of such location-specific data has resulted in Indian Meteorological Department (IMD) establishing two absolute thresholds to define heat waves for the entire nation (Indian Meteorological Department (IMD), 2018), which may not accurately capture the underlying population risk. The absolute $T_{\max}$ cutoffs are used to classify the heatwave events in India, where IMD weather stations' single-day $T_{\max} \geq 40^\circ\text{C}$ for plains and $\geq 30^\circ\text{C}$ for hilly regions are characterized as heatwaves (National Disaster Management Authority (NDMA), n.d.).

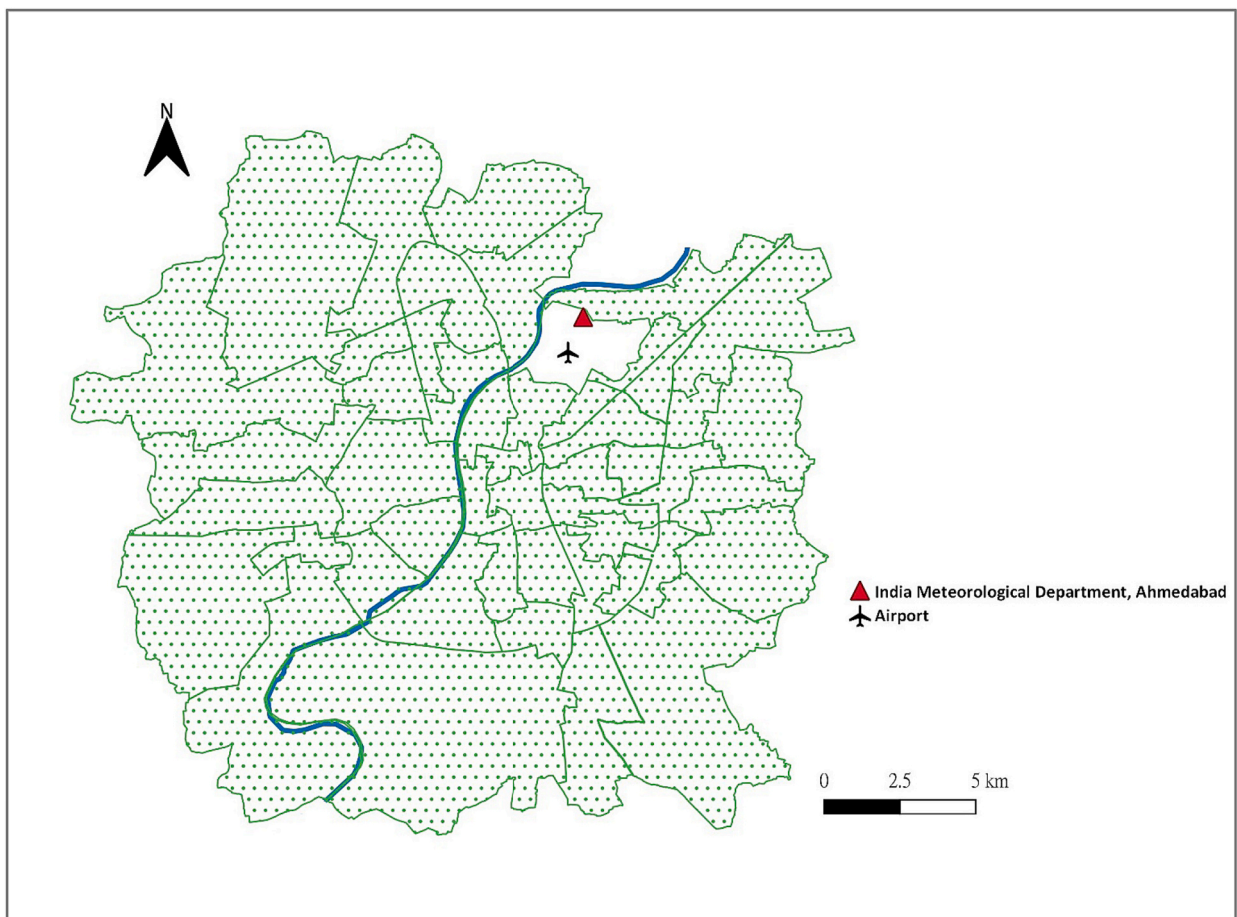


Fig. 1. Map of the study area showing local meteorological station and ward boundaries of Ahmedabad city.

If this criteria met at least in two stations in a meteorological sub-division for at least two consecutive days then heatwave is declared on the second day. Prior studies have suggested that definitions of heatwaves may have implications for human health impacts and, thus, must be explored locally (Robinson, 2001).

Therefore, in this study, we have focussed on quantifying the mortality risk associated with varying temperature thresholds to better understand the impact of EHEs on mortality in India. We have quantified the effects of EHEs in terms of intensity and duration. We have also assessed the temperature exposure-response curve for Ahmedabad (reflecting MMT) along with threshold-specific mortality risk. The region-specific exposure-response curve can help to understand human heat exchange ability, thermoregulation limits and adaptation to heat of the population at a local level (Follos et al., 2021). This study also evaluated the impact of extended periods of extreme temperatures on the population. The primary objective of this study is to pinpoint the most suitable temperature indicator and threshold for activating the city's HAP via a comprehensive comparison. Additionally, since various factors influence population response to heat, potential effects of gender and time-varying (lagged) exposures were also evaluated.

2. Methodology

2.1. Study area

Ahmedabad, a city in western India situated along the seasonal river Sabarmati as shown in Fig. 1, is characterized by a hot and arid climate, with summer temperature often exceeding 45 °C. Recent urban expansion and development have exacerbated the UHI effects in this region. The city has shown a growing incidence of heat-related fatalities. In response, the city's government enacted a HAP in 2013 (Dutta et al., 2021; Knowlton et al., 2014) to facilitate heatwave awareness and adaptation strategies among the population. This initiative has proven effective, potentially preventing an estimated annual average of 1190 deaths (Hess et al., 2018). Therefore, Ahmedabad serve as an ideal setting to examine the impact of EHEs on all-cause mortality, further supporting the existing HAP in the city.

2.2. Data sources

We obtained daily all-cause mortality data (including accidental deaths) from the Ahmedabad Municipal Corporation (AMC) office of the Registrar of Births and Deaths from 2002 to 2018. AMC collects daily mortality data for the entire city from all sources, including hospitals, crematoria, and burial grounds. All deaths within the city must be officially registered. Hospitals, both government and private, report deaths occurring within their premises. Deaths happening at home or elsewhere are reported by family heads to the clerks at crematoria or burial grounds, and this information is later forwarded to the municipal corporation's zone-specific offices. In total, 684,142 deaths were registered from 2002 to 2018 associated with all-cause mortality in Ahmedabad city of India. Overall, 60% of total deaths were reported among males (Table 1), with an overall increasing mortality trend among both gender during the study period (Fig. 2). The census population for the respective periods i.e. 2001 and 2011 was obtained from the Office of the Registrar General Census, Indian Census Bureau (Ahmedabad City Census, 2011).

The daily temperature records for the same period, including maximum temperature ($T_{\max}$), minimum temperature ($T_{\min}$), and average temperature (T_{avg}), were retrieved from ERA5 reanalysis (ECMWF Reanalysis 5th Generation) products. ERA5 provides hourly estimates of climate variables, detailed information on the ERA5 reanalysis product can be checked at <https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>. During the study period, the daily $T_{\max}$ in Ahmedabad city ranged from 15 °C to 45 °C (Table 1). In addition, monthly weather station data of Ahmedabad city was also obtained from the Indian Institute of Tropical Meteorology (IITM), Pune, to cross-check the accuracy of the ERA5 reanalysis product. Due to the limited data availability of daily Ahmedabad meteorological records, we used the ERA5 reanalysis product as a substitute for the weather station.

We identified that monthly $T_{\max}$ in the ERA5 dataset is comparable to weather station observations ($R^2 = 0.85$) and Global Historical Climatology Network (GHCN) data ($R^2 = 0.92$) as shown in Supplementary Fig. 1, the gridded spatial continuity of ERA5 can have advantages over in-situ measurements for regional modeling applications (Mistry et al., 2022; Urban et al., 2021). The station-based data exhibited a variance of approximately ~5 degrees in comparison to the ERA5 reanalysis. This variance can be attributed to the fact that ERA5 operates on a 0.25-degree (~25 km) grid, making it sensitive to local biases that a single weather station may not

Table 1

Daily number of all-cause mortality and weather variables from 2002 to 2018 in Ahmedabad, India.

	Total Deaths (%)	Average daily deaths/100,000	Mean	Min	P50	P95	Max
Total deaths	684,142 (100)	2.19	111	49	107	156	315
Male deaths	411,749 (60)	1.32	67	25	64	96	176
Female deaths	272,349 (40)	0.87	44	13	43	64	141
Weather Variables							
$T_{\max}$ (°C)	–	–	32	15	32	40	45
$T_{\min}$ (°C)	–	–	21	6	23	28	32
T_{avg} (°C)	–	–	26	13	27	33	37

Note: P50 and P95 represent the 50th and 95th percentile values.

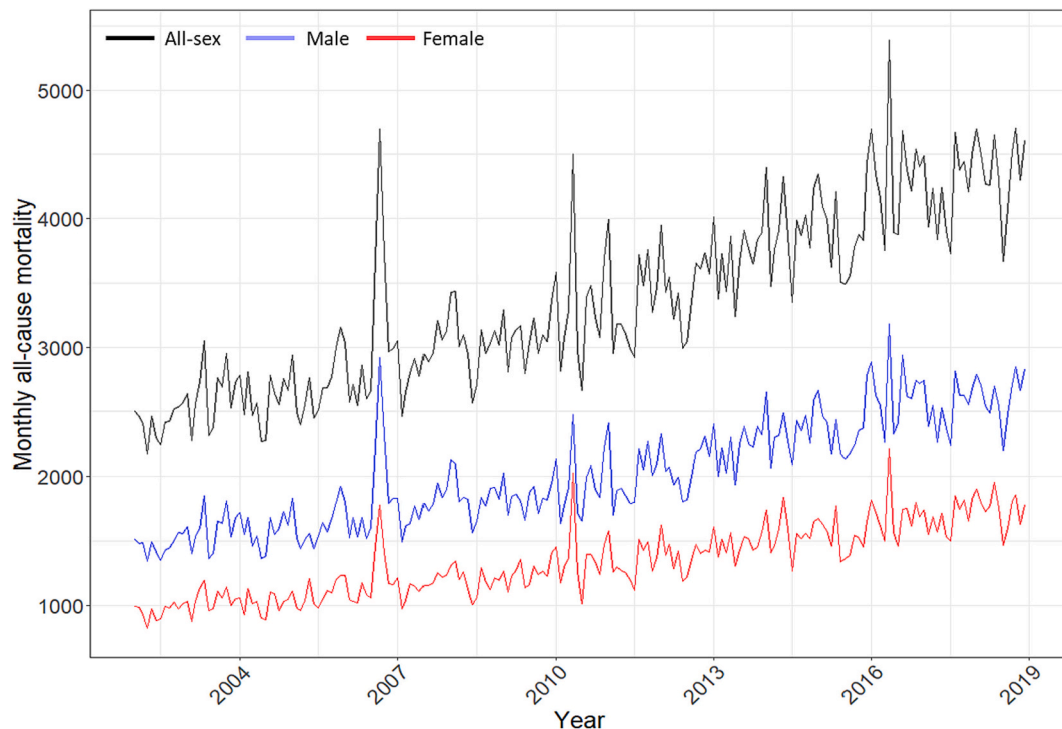


Fig. 2. Trend of monthly gender-stratified all-cause mortality from 2002 to 2018 in Ahmedabad, India.

accurately represent across the entire grid. Nevertheless, ERA5 can effectively capture the fine-scale temperature fluctuations within the city limits.

2.3. Classification of extreme heat events

We explored the effects of various EHEs thresholds (intensities and durations) on the mortality in Ahmedabad city. The reference point was kept at 40 °C or 95th percentile, as it is used by IMD and it triggers Ahmedabad city's HAP. The distribution of temperature metrics in the city was further explored to derive temperature thresholds as shown in Supplementary Fig. 2. Next, we considered temperature cut-offs, ranging from the 85th to the 99.5th percentiles to assess the impact of temperature extremes within these percentile ranges. In addition to relative thresholds, we investigated the effects of absolute temperature cutoffs, including 40 °C, 41 °C, and 45 °C.

To account for the duration of EHEs, we categorized them into three groups based on the number of consecutive days: 0 days, 2 days, and 3 days. This categorization enabled us to investigate how the effects of extreme temperatures varied across these different

Table 2

Classification of different extreme heat events (EHEs) in Ahmedabad city of, India, from 2002 to 2018.

EHEs definition	Duration of days	Number of events
$T_{\max} \geq 40^{\circ}\text{C}$ or $P95_{(0d)}$	0	266
$T_{\max} \geq 40^{\circ}\text{C}$ or $P95_{(2d)}$	2	78
$T_{\max} \geq 40^{\circ}\text{C}$ or $P95_{(3d)}$	3	31
$T_{\max} \geq 41^{\circ}\text{C}_{(0d)}$	0	164
$T_{\max} \geq 45^{\circ}\text{C}_{(0d)}$	0	2
$T_{\max} \geq P85_{(0d)}$	0	884
$T_{\max} \geq P85_{(2d)}$	2	555
$T_{\max} \geq P85_{(3d)}$	3	384
$T_{\max} \geq P90_{(0d)}$	0	574
$T_{\max} \geq P90_{(2d)}$	2	271
$T_{\max} \geq P90_{(3d)}$	3	90
$T_{\max} \geq P93_{(0d)}$	0	449
$T_{\max} \geq P97_{(0d)}$	0	198
$T_{\max} \geq P99_{(0d)}$	0	77
$T_{\max} \geq P99.5_{(0d)}$	0	33

Note: P85 to P99.5 represent the 85th to 99.5th percentile values.

timeframes. Since events with $T_{\max}$ exceeding the 97th or higher percentiles for extended durations were rare, we considered prolonged effects up to the 95th percentile for consecutive days. To manage the complexity of the analysis and mitigate potential issues related to multicollinearity and overlapping effects, we separately assessed the impact of EHEs intensities and durations on mortality. As a result, we conducted individual tests for each of the 12 EHE duration-based definitions listed in Table 2 within our model to gain a comprehensive understanding of their effects.

2.4. Statistical analysis

Mann Kendall test (at a 95% confidence interval) was used to understand the data trend over time, and the significance of the trend was estimated using the *Kendal* package in R software (version 3.4.0). A Distributed Lag Non-Linear Model (DLNM) was used to assess the temperature-associated mortality risk using *DLNM* package of R software (Gasparrinia et al., 2010). To enable the DLNM model to capture the non-linear and delayed temperature-mortality association, lag-stratified crossbasis function with covariates and poisson family was adopted. Within the temperature matrices, a basic spline with 7 degrees of freedom (*df*) was applied, and the knots were strategically placed at equally spaced quantiles of the predictor. For lag, the knots were set at evenly spaced values on the logarithmic scale of lags, with a maximum lag of 21 days utilized in our study. We extended the lag period to 21 days to encompass prolonged heat effects and to mitigate the influence of short-term effects (harvesting effect). This was also assessed through sensitivity analysis. We have reported the cumulative 21-day mortality risk along with its corresponding 95% confidence interval (CI) for specific EHEs thresholds. We incorporated natural cubic splines (*ns*) with 7 *df* per year to account for seasonal fluctuations in time. Holidays and days of the week were also included in the model as categorical variable. Additionally, EHEs durations (consecutive 0, 2 and 3 days) based categorical variables for different temperature thresholds were also individually incorporated into the model. The effects of these EHEs durations were estimated from the exponential of coefficients. The model equation is described below:

$$\text{Log}[Y] = \beta + cb(T, 7df, lag, 21) + ns(date, 7df \text{ per year}) + EHEs_{(d)} + DOW + Holiday + \log(Population)$$

Here, β is the model intercept; Y is the daily all-cause mortality or deaths; T is the exposure variable of temperature (including $T_{\max}$, $T_{\min}$, and T_{avg}) and *lag* is the lag of 21 days. In cross-basis (*cb*) matrix, basic spline (*bs*) function with seven degrees of freedom (*df*) was used for daily T . $EHEs_{(d)}$ are the duration based binary variables for specific thresholds of temperature. $EHEs_{(d)}$ takes on a value of 1 when extreme temperatures were recorded in the city for a specific number of consecutive days, while it holds a value of 0 for days when non-extreme temperatures were observed. A natural cubic spline (*ns*) with seven *df* was used to control the seasonal effects. Additionally, the population log was used as an offset variable to control population effects over mortalities. Initially, we employed the mean temperature as the central value for all temperature matrices to determine the optimal temperature, i.e., MMT. Subsequently, we utilized this MMT to re-center our exposure-response curves for temperature and mortality. Furthermore, we also calculated the attributable fractions (AF) associated with the respective definitions of EHEs using *attrdl* function of R package. We used “backward” function to calculate retrospective AFs with 95% CI using Monte Carlo simulations (Gasparrini and Leone, 2014).

2.5. Sensitivity analysis

We conducted a sensitivity analysis involving various temperature metrics, such as $T_{\max}$, T_{avg} , and $T_{\min}$. The goal was to identify the most suitable temperature indicator for our primary modeling. During this analysis, we examined the *df* for temperature-mortality curves, which ranged from 6 to 7. In our study, we considered lagged effects using crossbasis from 0 to 7 days, 0–10 days, and 0–21 days. Furthermore, to address seasonal variations throughout the study period, we evaluated different time smoothing approaches, including 4 and 7 *df* per year. The lowest Akaike Information Criterion (AIC) values of models were used as the selection

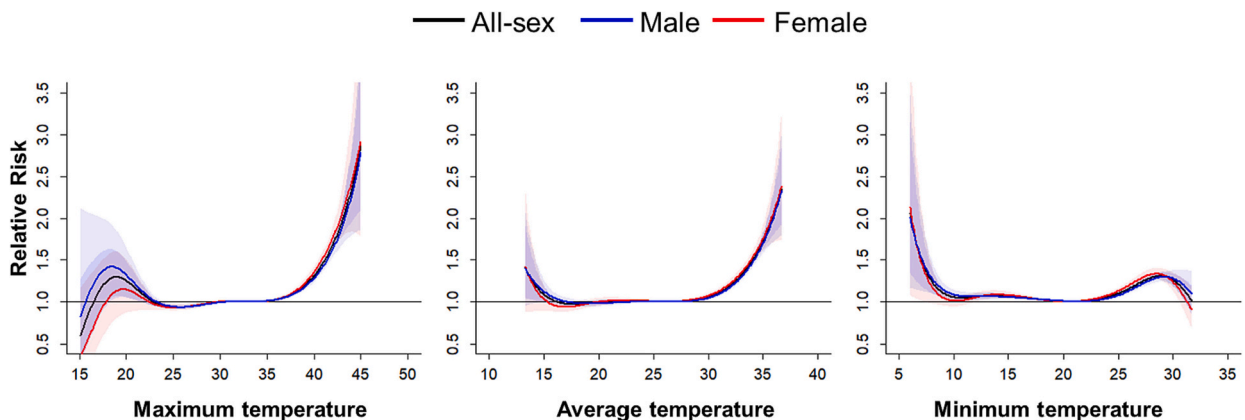


Fig. 3. Cumulative relative risk of all-cause mortality for lag of 21 days for different exposure matrices of temperature including: (a) Maximum temperature, (b) Average temperature and (c) Minimum temperature from 2002 to 2018 for Ahmedabad city of India compared with mean temperature of each matrix.

criteria.

3. Results

3.1. All-cause mortality and temperature characteristics

In recent years, Ahmedabad has observed a concerning uptrend in mortality rates. This was validated by the Mann-Kendall test, which demonstrated a significant positive mortality trend ($k\text{-tau} = 0.59$, $p\text{-value} < 0.001$) over the study period. Notably, when assessing the trends of monthly $T_{\max}$, $T_{\min}$, and T_{avg} between 2002 and 2018, it became evident that all temperature matrices are showing rising trend, particularly $T_{\min}$ exhibited a steeper increase in slope compared to $T_{\max}$ and T_{avg} . This observation could potentially be attributed to the pronounced UHI effect in the city (Supplementary Fig. 3).

3.2. Temperature-related mortality risks

To understand the rising mortality trend w.r.t. temperature, we have explored exposure-response curves of the all-age and all-sex population for all temperature matrices. These curves were constructed using the mean temperature as the reference point (Fig. 3). Since $T_{\max}$ revealed higher risk of all-cause mortality in the city it was used for the further modeling which was also supported by sensitivity analysis, as discussed in Section 3.4. The temperature-mortality curve of $T_{\max}$ revealed a typical 'J' shape suggesting elevated mortality risk on both sides of the temperature distribution. The curve was re-centered using the MMT of 26 °C (Fig. 4). MMT represents the temperature at which mortality reaches its minimum value. Thus, mortality attributable to heat is represented on the right side of MMT (red-colored), while cold-attributable mortality is represented to the left (blue-colored).

The temperature mortality relationship showed a higher relative risk of all-cause mortality with $T_{\max} > 35$ °C. Table 3 presents gender-stratified mortality risks for relative and absolute $T_{\max}$ thresholds and Fig. 5 shows the exposure-response curves to provide comparative evidence. The findings suggested that mortality risk for all-age-sex-specific populations increased by 13% when using the 85th percentile temperature threshold (RR:1.13; 95% CI: 1.10–1.16). Further, the mortality risk increased to RR:1.19 (95% CI: 1.16–1.23) and RR:1.74 (95% CI: 1.66–1.82) when using the extreme temperature threshold of 90th and 99.5th percentile, respectively. Likewise, using a single-day absolute $T_{\max}$ above 40 °C as the cut-off threshold resulted in a 30% increase in mortality risk (RR:1.30; 95% CI: 1.26–1.35). As expected, the highest risk was observed when we used 45 °C as the threshold (RR:3.08; 95% CI: 2.47–3.83).

Analysis stratified by gender showed higher EHE-related mortality risk among females than males (Table 3). For example, when $T_{\max}$ exceeded 40 °C, mortality risk among males increased by 25% (RR:1.25; 95% CI: 1.19–1.30), compared to a 40% increase in risk among females (RR:1.40; 95% CI: 1.32–1.48). Additionally, the mortality risk during severe heatwave events ($T_{\max} \geq 45$ °C) was

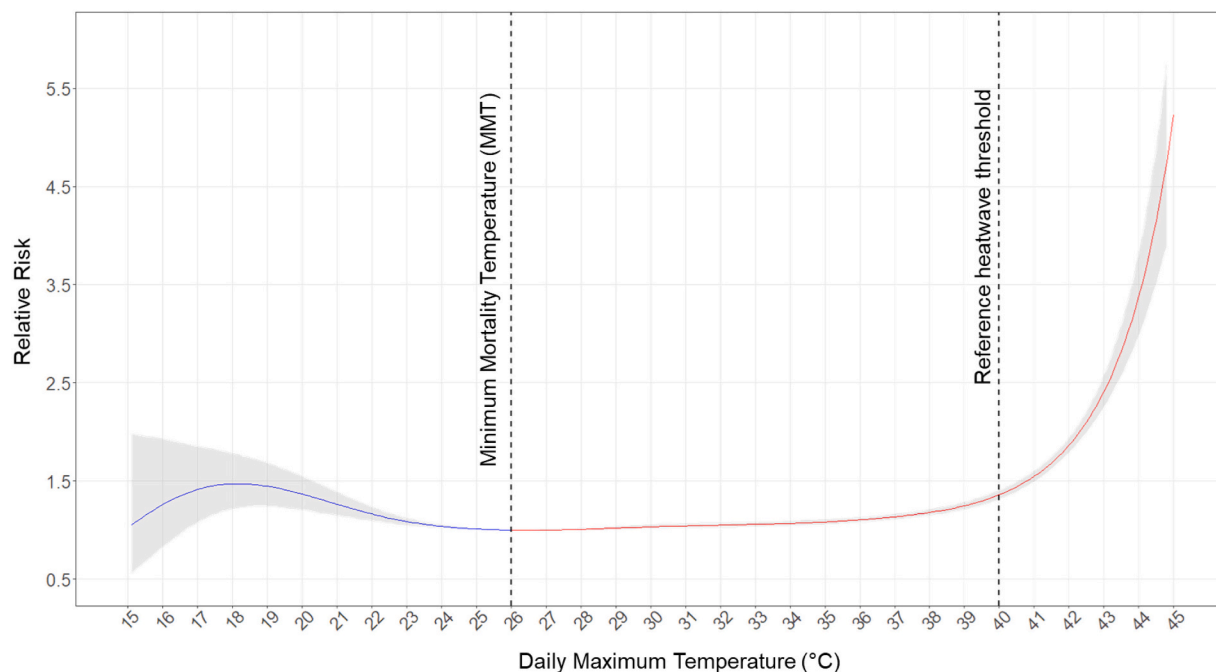


Fig. 4. Temperature–mortality relationship for lag 21 days from 2002 to 2018. Dotted vertical line shows the minimum mortality temperature (MMT) and reference heatwave threshold for Ahmedabad.

Table 3

Cumulative relative risk (at 95% CI) of all-cause mortality for lag of 21 days at specific thresholds of temperature extremes ($T_{\max}$) in Ahmedabad, India, from 2002 to 2018.

EHEs intensities	All-sex RR (95% CI)	Male RR (95% CI)	Female RR (95% CI)
$T_{\max} = 40^{\circ}\text{C}$ or P95	1.30 (1.26, 1.35)^a	1.25 (1.19, 1.30)	1.40 (1.32, 1.48)
$T_{\max} = 41^{\circ}\text{C}$	1.45 (1.40, 1.50)	1.37 (1.31, 1.44)	1.57 (1.48, 1.66)
$T_{\max} = 45^{\circ}\text{C}$	3.08 (2.47, 3.83)	3.03 (2.28, 4.02)	3.11 (2.20, 4.38)
P85 (38.0 $^{\circ}\text{C}$)	1.13 (1.10, 1.16)	1.10 (1.06, 1.14)	1.18 (1.13, 1.23)
P90 (38.9 $^{\circ}\text{C}$)	1.19 (1.16, 1.23)	1.15 (1.11, 1.20)	1.26 (1.20, 1.32)
P93 (39.5 $^{\circ}\text{C}$)	1.25 (1.21, 1.29)	1.20 (1.15, 1.25)	1.33 (1.26, 1.40)
P97 (40.7 $^{\circ}\text{C}$)	1.40 (1.35, 1.45)	1.33 (1.27, 1.39)	1.51 (1.43, 1.60)
P99 (41.8 $^{\circ}\text{C}$)	1.61 (1.55, 1.68)	1.52 (1.44, 1.60)	1.75 (1.64, 1.87)
P99.5 (42.3 $^{\circ}\text{C}$)	1.74 (1.66, 1.82)	1.64 (1.54, 1.75)	1.89 (1.75, 2.04)

Note: P85 to P99.5 represent the 85th and 99.5th percentile values.

^a Bold font indicates statistical significance.

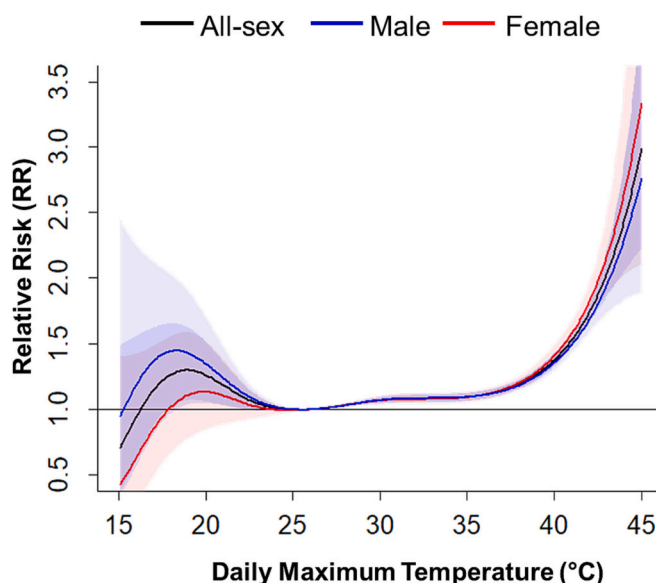


Fig. 5. Cummulative relative risk of all-cause mortality in association with $T_{\max}$ by gender for lag of 21 days from 2002 to 2018 in Ahmedabad city of India, compared to reference minimum mortality temperature (MMT) i.e. 26°C .

considerably higher among females (RR: 3.11; 95% CI: 2.20–4.38) compared to males (RR:3.03; 95% CI: 2.08–4.02).

3.3. Impact of duration and lags on temperature extremes

The examination of consecutive days of extreme temperature thresholds revealed distinct associations with different thresholds. Notably, the existing threshold for heat alerts in the city displayed no significant effects related to duration (Table 4). In contrast, the 85th percentile exhibited significant effects on the same day of exposure, whereas consecutive days of extreme temperatures did not

Table 4

Relative risk (at 95% CI) of all-cause mortality in association with consecutive days of $T_{\max}$ extremes in Ahmedabad, India, from 2002 to 2018.

EHEs duration	All-sex RR (95% CI)	Male RR (95% CI)	Female RR (95% CI)
$T_{\max} \geq \text{P85}_{(0d)}$	1.02 (1.01, 1.04)^a	1.02 (1.00, 1.04)	1.03 (1.01, 1.06)
$T_{\max} \geq \text{P85}_{(2d)}$	0.99 (0.98, 1.00)	1.00 (0.98, 1.02)	0.98 (0.96, 1.00)
$T_{\max} \geq \text{P85}_{(3d)}$	0.99 (0.97, 1.00)	0.99 (0.98, 1.01)	0.97 (0.95, 0.99)
$T_{\max} \geq \text{P90}_{(0d)}$	1.03 (1.02, 1.05)	1.04 (1.02, 1.06)	1.03 (1.00, 1.05)
$T_{\max} \geq \text{P90}_{(2d)}$	1.00 (0.99, 1.02)	1.00 (0.99, 1.02)	1.00 (0.97, 1.01)
$T_{\max} \geq \text{P90}_{(3d)}$	1.00 (0.98, 1.02)	1.01 (0.99, 1.03)	0.99 (0.96, 1.01)
$T_{\max} \geq \text{P95}_{(0d)}$	1.03 (1.01, 1.05)	1.02 (1.00, 1.05)	1.04 (1.01, 1.07)
$T_{\max} \geq \text{P95}_{(2d)}$	1.00 (0.98, 1.02)	1.00 (0.98, 1.04)	0.98 (0.95, 1.02)
$T_{\max} \geq \text{P95}_{(3d)}$	0.99 (0.95, 1.02)	1.02 (0.97, 1.06)	0.94 (0.89, 0.99)

^a Bold font indicates statistical significance.

demonstrate any notable associations. Likewise, the 95th percentile indicated significant effects on the same day of exposure, with a reduction in effects observed for consecutive days of the same temperature.

Similarly, lag-specific analysis (Fig. 6) also showed the mortality risk to be highest during the day of exposure (lag 0), with the risk decreasing with increasing lag time. Significant increases in mortality risks were identified up to a lag of two days at lower percentiles of $T_{\max}$. However, higher thresholds (relative and absolute) resulted in a more sustained lag effect. For example, 40 °C as a cut-off threshold resulted in increased mortality risk at lag0 and lag1, while using 45 °C as a threshold resulted in increased mortality risk from lag 0 through lag 7 days. A significantly increased mortality risk was also identified after a lag of 16 days with 45 °C as a threshold, which may be an effect of long-term and continuous exposure to extreme heat days.

3.4. Attributable fractions

Table 5 displays the AFs linked to various EHE definitions in the city of Ahmedabad. Notably, the most substantial AFs, accounting for 3.58% of total mortalities (95% CI: 3.20–3.96), were associated with $T_{\max}$ at or above the 85th percentile, equivalent to 38 °C. In contrast, the existing IMD threshold, set at temperatures of ≥ 40 °C or $\geq$ 95th percentile of $T_{\max}$, contributed to 1.96% of total mortalities in the city (95% CI: 1.80–2.11). The gender-based comparison revealed that AFs were higher among females compared to males. It is noteworthy that the city experienced a lower fraction of deaths during high-intensity EHEs when compared to RRs at these levels, due to rare occurrence of such events.

The sensitivity analysis of various lag days (0–7, 0–10, and 0–21 days), showed the lowest AIC with lag of 21 days for gender groups. While, for the all-sex category, a lag of 10 days reported a lower AIC (Supplementary Table 1). The variation in AIC values between a 10-day lag and a 21-day lag was not substantial. Thus, we opted to use a 21-day lag for the final modeling. Furthermore, the exposure-response curves also did not exhibit significant variation for different lag days across different temperature matrices (Supplementary Fig. 4). Additionally, the comparison of all three temperature matrices revealed best performance of $T_{\max}$. The selection of the final smoothing functions for the $T_{\max}$ matrix was also based on the lowest AIC values (Supplementary Table 2).

4. Discussion

Previous studies have extensively documented the impact of extreme heat exposure on mortality and morbidity, but mostly from the western world with developed economy. However, there have been limited studies in the global south that investigate the impacts of EHEs across various intensities, durations and prolonged temperature extremes. There are several important factors are which less understood in terms of temperature extremes, such as how the local population adapts to these extreme heat exposure, how different thresholds used to define these extreme events impact observed association, and whether or not the timing of exposure has an implication on observed health burden.

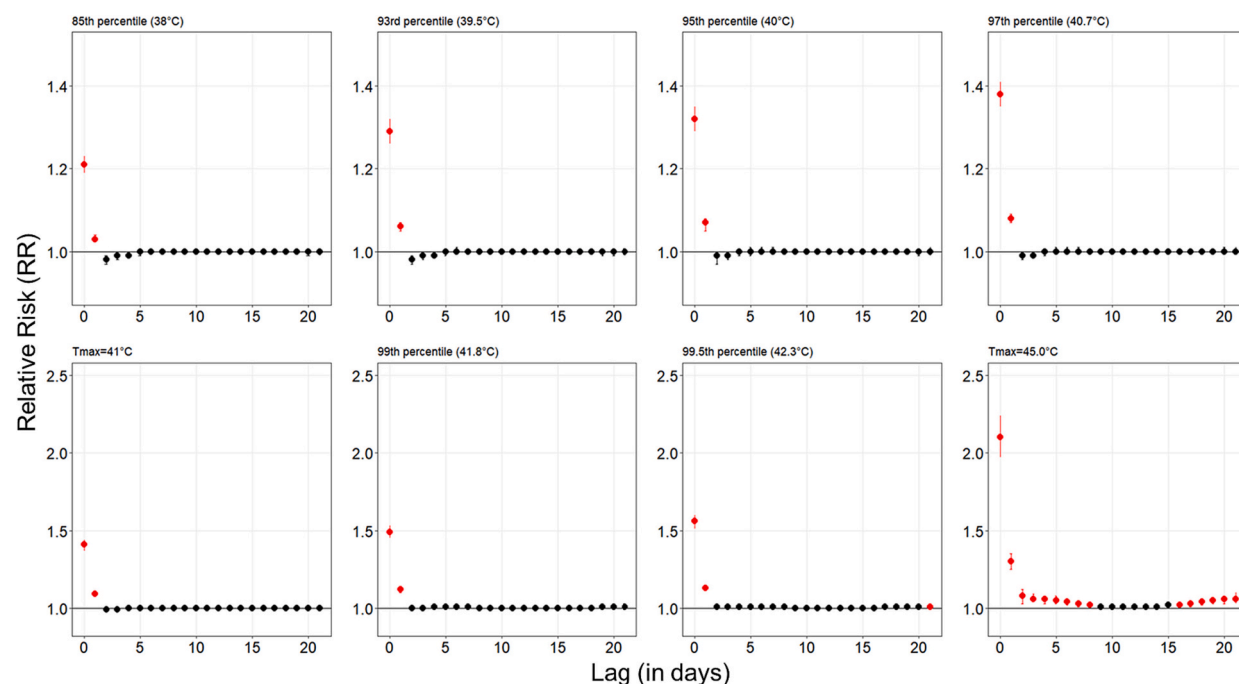


Fig. 6. Lag-specific effect on the relative risk (RR) (at 95% CI) of all-cause mortality at different temperature thresholds in Ahmedabad from 2002 to 2018 [Red colored error bars represent significant positive RR, while the black represents non-significant RR]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Attributable fractions (AFs) of daily all-cause mortality in association with different intensities of extreme heat events (EHEs) on the same day of exposure in Ahmedabad, India, from 2002 to 2018.

EHEs intensities	All-sex AF (95% CI)	Male AF (95% CI)	Female AF (95% CI)
$T_{\max} \geq 40^{\circ}\text{C}$ or $\geq P95_{(od)}$	1.96 (1.80, 2.11)^a	1.68 (1.48, 1.88)	2.37 (2.12, 2.62)
$T_{\max} \geq 41^{\circ}\text{C}_{(od)}$	1.28 (1.19, 1.38)	1.12 (0.99, 1.24)	1.53 (1.38, 1.68)
$T_{\max} \geq 45^{\circ}\text{C}_{(od)}$	0.04 (0.03, 0.04)	0.04 (0.03, 0.04)	0.04 (0.03, 0.05)
$\geq P85$ (38.0 $^{\circ}\text{C}$) _(od)	3.58 (3.20, 3.96)	3.00 (2.47, 3.51)	4.47 (3.86, 5.06)
$\geq P90$ (38.9 $^{\circ}\text{C}$) _(od)	2.91 (2.63, 3.19)	2.46 (2.09, 2.81)	3.60 (3.16, 4.03)
$\geq P93$ (39.5 $^{\circ}\text{C}$) _(od)	2.35 (2.15, 2.55)	2.01 (1.74, 2.26)	2.88 (2.57, 3.19)
$\geq P97$ (40.7 $^{\circ}\text{C}$) _(od)	0.72 (0.66, 0.78)	1.27 (1.13, 1.41)	1.76 (1.59, 1.93)
$\geq P99$ (41.8 $^{\circ}\text{C}$) _(od)	1.73 (1.67, 1.80)	0.63 (0.56, 0.71)	0.85 (0.76, 0.94)
$\geq P99.5$ (42.3 $^{\circ}\text{C}$) _(od)	0.37 (0.34, 0.41)	0.33 (0.29, 0.38)	0.43 (0.37, 0.49)

^a Bold font indicates statistical significance.

The present study has used long historical records (2002–2018) of daily mortality data (684 k deaths) recorded in Ahmedabad city. In the context of temperature matrix selection for issuing warnings, our study offers empirical insights from an Indian standpoint. It emphasizes the effectiveness of $T_{\max}$ in representing associated health risks, particularly in the dry and arid regions of the country. Moreover, our study discovered that even relatively modest temperature thresholds, as low as the 85th percentile, displayed a robust association with mortality within the context of EHEs. Notably, our findings differ from what has been commonly reported in the literature, it revealed immediate effects of heat and consecutive days of EHEs did not exhibit a significant association with mortality risks. The impact of intensity in EHE classification significantly outweighs the additional influence of duration. Additionally, we observed that higher thresholds resulted in more sustained risk, lasting up to seven days after the exposure compared to the day observed for lower thresholds. Irrespective of the thresholds used, females appeared to be at higher risk of death from exposure to EHEs.

While many regions typically observe U-shaped temperature-mortality curves, signifying increased risks during both heat and cold extremes (Gasparrini et al., 2015; Lin et al., 2020; Bao et al., 2016; Guo et al., 2014), our study in Ahmedabad revealed a subtler impact of cold compared to the pronounced heat effects (J-shaped curve). This observation aligns with the findings of a similar study carried out in Delhi, India, which reported higher mortality risks associated with heat compared to cold (McMichael et al., 2008). Additionally, our analysis has pinpointed a MMT of 26 $^{\circ}\text{C}$ for Ahmedabad. Beyond this threshold, there is a notable increase in heat-related risk, particularly exceeding 35 $^{\circ}\text{C}$, which approximately corresponds to the 70th percentile of temperatures. A multicity international study has also indicated a similar range with heat-related deaths occurring between a temperature of 16 $^{\circ}\text{C}$ to 31 $^{\circ}\text{C}$ (McMichael et al., 2008). Furthermore, another study from Taiwan, a region with tropical to sub-tropical climate, where the exposure matrix employed was the average temperature also reported a MMT of 26 $^{\circ}\text{C}$ (Lin et al., 2011). Comparable temperature ranges of MMT has been suggested by Gasparrini et al. as well with a percentile of around 60th for tropical areas and a range between the 80th and 90th percentile for temperate regions (Gasparrini et al., 2015).

These findings raise an important question: Are the currently defined temperature thresholds responsive to population health risks, particularly in cities characterized by hot summers, such as Ahmedabad? Further research and consideration are warranted to address this question. Even though a greater risk was identified to higher intensities of EHEs, low intensities also considerably increased the mortality risk among our study population. Similar findings have been reported from US, they identified two-fold increase in heat-related RR at >99.5th percentile compared to 98th percentile (Anderson and Bell, 2009). Even though it won't be a new finding to say that higher intensities of EHEs increased the risk of mortality, however, it's understanding from a location-specific context is important.

Furthermore, evaluating the impact of EHEs on mortality involves considering their intensity, duration, and timing (Lin et al., 2011). It's worth noting that the impact of high temperatures is primarily observed over the short term (Chen et al., 2018). Although studies from temperate regions have consistently shown that long-duration heat waves are linked to increased mortality risks (Montero et al., 2012; Sheridan and Lin, 2014). However, the definition of prolonged heat events varies across studies, and many of these studies focused on single or isolated heat waves, which may not represent the typical scenarios experienced by the population (Ho et al., 2017). In regions with tropical to sub-tropical climates, particularly those with hot summers, immediate effects of temperature extremes are more prominent than delayed effects (Sharma et al., 2023). For instance, a study in Hong Kong, a sub-tropical high-density city, reported that consecutive hot nights were associated with higher mortality risks, while consecutive hot days did not exhibit a significant connection with excess mortality. Additionally, non-consecutive hot days contributed to mortality risk (Ho et al., 2017).

Moreover, our findings align with those of Gasparrini and colleagues, who observed that temperature intensity has a more pronounced effect on mortality risk than prolonged exposure to extreme temperatures over consecutive days (Gasparrini and Armstrong, 2011). Another study noted that risk estimates using the DLNM model with a lag of 0 days were slightly higher than those with a 30-day lag (Lin et al., 2011). In Ahmedabad city, the highest risk was observed on the day of exposure, consistent with findings in Varanasi, India (Singh et al., 2019). The immediate temperature effects in tropical regions may be attributed to the more frequent and prolonged exposure of their populations to high temperatures compared to regions with temperate climates (Kim et al., 2019). Further research is needed to delve into the specific mechanisms underlying this observation. Nevertheless, these insights underscore the diverse temperature-mortality associations in different climate zones.

We also observed higher EHE-related mortality risk among women compared to men. This finding is consistent with other studies of

heatwave-related deaths that have reported higher risk among females than males (Singh et al., 2019; Lung et al., 2021; Faye et al., 2021; Tong et al., 2015). Similarly, a previous study focusing on Ahmedabad's May 2010 Heatwave noted higher female death rates (Azhar et al., 2014). Previous studies have hypothesized that females' sweat glands are activated at higher temperatures than males causing less heat exchange between the body and environment, which may put women at a higher risk of heat-related mortality (Kuchcik, 2021; Hajat et al., 2007; Rey et al., 2009). Besides, social and cultural norms, including the higher burden of household activities and culturally influenced dressing practices resulting in a higher exposure propensity among women, may explain some of the higher risks (McGregor et al., 2015).

Although it might be commonly assumed that heat affects everyone equally, our study provides evidence that women face a higher risk of heat-related deaths. These findings demand attention particularly for women in developing nations where poverty, social and cultural norms and gender inequity is deeply rooted. The socio-cultural practice, as in many developing countries expect women to stay indoors for household chores which may increase their exposure to indoor heating. To a considerable extent, cooking areas in many Indian households are poorly ventilated (Padhi and Padhy, 2008) and women spend much of their time indoors in cooking areas which exposes them to higher heat effects. Thus, these findings strongly suggest policymakers in developing countries must consider gender-specific heatwave prevention plans. Moreover, low cost household cooling strategies can be lifesaving (Sharma et al., 2022).

Even though this study addressed crucial information about temperature-associated mortality risk, several shortcomings are noted. For instance, the study did not consider individual temperature exposure for evaluation. Likewise, we did not estimate age-specific effects due to data limitations. The mortality rates calculated in the study assumed that population growth remained stagnant for 10 years and utilized census-based population data, updated once per decade, as the denominator. All-cause mortality data that we used in this study (including accidental deaths) for heat-health analysis are generally reliable. We did not use cause-specific mortality data because this information was not available and for India per se, there are several potential issues with cause-specific mortality data; firstly, reporting conventions are not uniform across the country making comparisons between different cities difficult, and secondly, in some regions, the causes of death are not accurately reported if it is case sensitive. In addition, some of the deaths may go unreported, resulting in an underestimation. Moreover, underlying comorbidities could not be accounted for, as such information was unavailable. Likewise, we did not account for other environmental exposures, including air pollution, relative humidity, wind speed etc. Finally, the station-based data exhibited a variance of approximately ~ 5 degrees in comparison to the ERA5 reanalysis, which can be due to 0.25-degree (~ 25 km) grid, making it sensitive to local biases that a single weather station may not accurately represent across the entire grid. However this did not differ between heat wave vs non heat wave period, so the potential exposure misclassification is non-differential.

Despite these potential concerns, applying these results is crucial to fill the knowledge gap in heat alert systems and policymaking. Identifying local MMT or an optimum temperature threshold can intensify heatwave prevention measures. With the increasing frequency, duration and intensity of extreme weather events, future heat action plans and management guidelines require consideration of EHEs to minimize the societal and life losses associated with temperature rise. The findings of this study can benefit local government, the health sector, and urban planners to develop effective area-specific integrated risk management measures for reducing heat-associated fatalities. Moreover, the health risk of various temperature intensities or EHEs can facilitate a cost-effective heatwave early warning system. Our findings emphasize the need to broaden public health policies and adaptation strategies to address the full spectrum of temperature-related EHE effects. While further research is required to determine the preventable portion of excess mortality for each component, it is clear that timely preventive measures are crucial for mitigating the health impacts of high temperatures.

5. Conclusion

There is an urgent need to develop local-level heat action plans and early warning systems to enhance community resilience to the threats of climate change-driven extreme heat events. An in-depth understanding of location-specific minimum mortality temperature and the implication of using different thresholds on actual disease burden will help inform appropriate heat action plans and early warning systems. Our data show that magnitude of the risk is more robustly related to the intensity used to identify EHEs compared to duration. Likewise, higher threshold results in more sustained risk that exceeds the day of exposure. Additional studies are needed to understand cause-specific death associated with high vs. low-intensity EHEs and sustained elevated risk.

Fundings

This study is funded by the Taiwan Ministry of Science and Technology (MOST 108-2625-M-033-002, MOST 109-2621-M-033-001-MY3, MOST 109-2625-M-033-002, and MOST 110-2625-M-033-002). Grants from the National Science Foundation and Swedish Research Council for Health, Working Life and Welfare (Forte) (project 2019-01552) provided further support through Belmont Forum (Award Number (FAIN): 2025470). In addition, we are also thankful to Academia Sinica (AS-SS-111-03) for their extended financial support.

CRedit authorship contribution statement

Ayushi Sharma: Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Priya Dutta:** Data curation, Writing – review & editing. **Priyanka Shah:** Data curation, Writing – review & editing. **Veena Iyer:** Data curation, Writing – review & editing. **Hao He:** Data curation, Writing – review & editing. **Amir Sapkota:** Funding acquisition, Writing – review & editing.

Chuansi Gao: Funding acquisition, Writing – review & editing. **Yu-Chun Wang:** Conceptualization, Funding acquisition, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

The data that has been used is confidential.

Acknowledgments

We are highly grateful to the Ahmedabad Municipal Corporation (AMC) office of the Registrar of Births and Deaths for sharing the all-cause mortality data of the city. We would also like to thank the study funders, including the Ministry of Science and Technology (MOST) of Taiwan, National Science Foundation, Swedish Research Council for Health, Working Life and Welfare and Academia Sinica. The work would not have been possible without their financial support.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.uclim.2024.101832>.

References

- Ahmedabad City Census. data Available online: <https://www.census2011.co.in/census/city/314-ahmedabad.html> (accessed on Oct 25, 2019).
- Anderson, B.G., Bell, M.L., 2009. Weather-related mortality: how heat, cold, and heat waves affect mortality in the United States. *Epidemiology* (Cambridge, Mass.) 20, 205.
- Azhar, G., 2019. Indian summer: three essays on heatwave vulnerability, estimation and adaptation. In: *Indian Summer: Three Essays on Heatwave Vulnerability, Estimation and Adaptation*. <https://doi.org/10.7249/rgsd431>.
- Azhar, G.S., Mavalankar, D., Nori-Sarma, A., Rajiva, A., Dutta, P., Jaiswal, A., Sheffield, P., Knowlton, K., Hess, J.J., 2014. Heat-related mortality in India: excess all-cause mortality associated with the 2010 Ahmedabad heat wave. *PLoS One* 9, e91831.
- Bao, J., Wang, Z., Yu, C., Li, X., 2016. The influence of temperature on mortality and its lag effect: a study in four Chinese cities with different latitudes. *BMC Public Health* 16, 375. <https://doi.org/10.1186/s12889-016-3031-z>.
- Chen, R., Yin, P., Wang, L., Liu, C., Niu, Y., Wang, W., Jiang, Y., Liu, J., Qi, J., 2018. Association between ambient temperature and mortality risk and burden: time series study in 272 main Chinese cities. *Bmj* 363.
- Dimitrova, A., Ingole, V., Basagaña, X., Ranzani, O., Milà, C., Ballester, J., Tonne, C., 2021. Association between ambient temperature and heat waves with mortality in South Asia: a systematic review and meta-analysis. *Environ. Int.* 146, 106170 <https://doi.org/10.1016/j.envint.2020.106170>.
- Dutta, P., Sathish, L., Mavankar, D., Ganguly, P.S., Saunik, S., 2020. Extreme heat kills even in very hot cities: evidence from Nagpur, India. *Int. J. Occup. Environ. Med.* 11, 188.
- Dutta, P., Golechha, M., Mavalankar, D., 2021. Prioritising climate change and health: the current status of heat-health in India. *Clim. Change Health Sector* 106–115.
- Faye, M., Dème, A., Diongue, A.K., Diouf, I., 2021. Impact of different heat wave definitions on daily mortality in Bandafassi, Senegal. *PLoS One* 16, e0249199.
- Follos, F., Linares, C., López-Bueno, J.A., Navas, M.A., Culqui, D., Vellón, J.M., Luna, M.Y., Sánchez-Martínez, G., Díaz, J., 2021. Evolution of the minimum mortality temperature (1983–2018): is Spain adapting to heat? *Sci. Total Environ.* 784, 147233 <https://doi.org/10.1016/j.scitotenv.2021.147233>.
- Fouillet, A., Rey, G., Laurent, F., Pavillon, G., Bellec, S., Guihenneuc-Jouyau, C., Clavel, J., Jougla, E., Hémon, D., 2006. Excess mortality related to the august 2003 heat wave in France. *Int. Arch. Occup. Environ. Health* 80, 16–24. <https://doi.org/10.1007/s00420-006-0089-4>.
- Gasparrini, A., Armstrong, B., 2011. The impact of heat waves on mortality. *Epidemiology* (Cambridge, Mass.) 22, 68.
- Gasparrini, A., Leone, M., 2014. Attributable risk from distributed lag models. *BMC Med. Res. Methodol.* 14, 55. <https://doi.org/10.1186/1471-2288-14-55>.
- Gasparrini, A., Guo, Y., Hashizume, M., Lavigne, E., Zanobetti, A., Schwartz, J., Tobias, A., Tong, S., Rocklöv, J., Forsberg, B., et al., 2015. Mortality risk attributable to high and low ambient temperature: a multicountry observational study. *Lancet* 386, 369–375. [https://doi.org/10.1016/S0140-6736\(14\)62114-0](https://doi.org/10.1016/S0140-6736(14)62114-0).
- Gasparrini, A., Armstrong, B., Kenward, M.G., 2010. Distributed lag non-linear models. *Stat. Med.* 29, 2224–2234. <https://doi.org/10.1002/sim.3940>.
- Guo, Y., Gasparrini, A., Armstrong, B., Li, S., Tawatsupa, B., Tobias, A., Lavigne, E., de Coelho, M., Leone, M., Pan, X., 2014. Global variation in the effects of ambient temperature on mortality: a systematic evaluation. *Epidemiology* (Cambridge, Mass.) 25, 781.
- Guo, Y., Gasparrini, A., Li, S., Sera, F., Vicedo-Cabrera, A.M., De Sousa Zanotti Stagliorio Coelho, M., Saldiva, P.H.N., Lavigne, E., Tawatsupa, B., Punnasiri, K., 2018. Quantifying excess deaths related to heatwaves under climate change scenarios: a multicountry time series modelling study. *PLoS Med.* 15, e1002629.
- Hajat, S., Kovats, R.S., Lachowycz, K., 2007. Heat-related and cold-related deaths in England and Wales: who is at risk? *Occup. Environ. Med.* 64, 93–100.
- Hashizume, M., Wagatsuma, Y., Hayashi, T., Saha, S.K., Streathfield, K., Yunus, M., 2009. The effect of temperature on mortality in rural Bangladesh—a population-based time-series study. *Int. J. Epidemiol.* 38, 1689–1697. <https://doi.org/10.1093/ije/dyn376>.
- Heaviside, C., Macintyre, H., Vardoulakis, S., 2017. The urban heat island: implications for health in a changing environment. *Curr. Environ. Health Rep.* 4, 296–305.
- Hess, J.J., Lm, S., Knowlton, K., Saha, S., Dutta, P., Ganguly, P., Tiwari, A., Jaiswal, A., Sheffield, P., Sarkar, J., 2018. Building resilience to climate change: pilot evaluation of the impact of India's first heat action plan on all-cause mortality. *J. Environ. Public Health* 2018.
- Ho, H.C., Lau, K.K.-L., Ren, C., Ng, E., 2017. Characterizing prolonged heat effects on mortality in a sub-tropical high-density city, Hong Kong. *Int. J. Biometeorol.* 61, 1935–1944. <https://doi.org/10.1007/s00484-017-1383-4>.
- India Meteorological Department, 2018. *Indian Meteorological Department (IMD)*, 3, 1–119.
- Kim, Y., Kim, H., Gasparrini, A., Armstrong, B., Honda, Y., Chung, Y., Ng, C.F.S., Tobias, A., Íñiguez, C., Lavigne, E., 2019. Suicide and ambient temperature: a multi-country multi-city study. *Environ. Health Perspect.* 127, 117007.
- Knowlton, K., Rotkin-Ellman, M., King, G., Margolis, H.G., Smith, D., Solomon, G., Trent, R., English, P., 2009. The 2006 California heat wave: impacts on hospitalizations and emergency department visits. *Environ. Health Perspect.* 117, 61–67.
- Knowlton, K., Kulkarni, S.P., Azhar, G.S., Mavalankar, D., Jaiswal, A., Connolly, M., Nori-Sarma, A., Rajiva, A., Dutta, P., Deol, B., et al., 2014. Development and implementation of South Asia's first heat-health action plan in Ahmedabad (Gujarat, India). *Int. J. Environ. Res. Public Health* 11.

- Kuchcik, M., 2021. Mortality and thermal environment (UTCI) in Poland-long-term, multi-city study. *Int. J. Biometeorol.* 65, 1529–1541. <https://doi.org/10.1007/s00484-020-01995-w>.
- Lin, Y.-K., Ho, T.-J., Wang, Y.-C., 2011. Mortality risk associated with temperature and prolonged temperature extremes in elderly populations in Taiwan. *Environ. Res.* 111, 1156–1163.
- Lin, Y.-K., Sung, F.-C., Honda, Y., Chen, Y.-J., Wang, Y.-C., 2020. Comparative assessments of mortality from and morbidity of circulatory diseases in association with extreme temperatures. *Sci. Total Environ.* 723, 138012 <https://doi.org/10.1016/j.scitotenv.2020.138012>.
- López-Bueno, J.A., Díaz, J., Follos, F., Vellón, J.M., Navas, M.A., Culqui, D., Luna, M.Y., Sánchez-Martínez, G., Linares, C., 2021. Evolution of the threshold temperature definition of a heat wave vs. evolution of the minimum mortality temperature: a case study in Spain during the 1983–2018 period. *Environmental sciences. Europe* 33, 101. <https://doi.org/10.1186/s12302-021-00542-7>.
- Lung, S.-C.C., Yeh, J.-C.J., Hwang, J.-S., 2021. Selecting thresholds of heat-warning systems with substantial enhancement of essential population health outcomes for facilitating implementation. *Int. J. Environ. Res. Public Health* 18, 9506.
- McGregor, G.R., Bessmoulin, P., Ebi, K., Menne, B., 2015. Heatwaves and health: guidance on warning-system development. WMOP (ISBN 9263111421.).
- McMichael, A.J., Wilkinson, P., Kovats, R.S., Pattenden, S., Hajat, S., Armstrong, B., Vajanapoom, N., Niciu, E.M., Mahomed, H., Kingkeow, C., et al., 2008. International study of temperature, heat and urban mortality: the 'ISOTHURM' project. *Int. J. Epidemiol.* 37, 1121–1131. <https://doi.org/10.1093/ije/dyn086>.
- Mistry, M.N., Schneider, R., Masselot, P., Royé, D., Armstrong, B., Kyselý, J., Orru, H., Sera, F., Tong, S., Lavigne, É., et al., 2022. Comparison of weather station and climate reanalysis data for modelling temperature-related mortality. *Sci. Rep.* 12, 5178. <https://doi.org/10.1038/s41598-022-09049-4>.
- Montero, J.C., Mirón, I.J., Criado-Álvarez, J.J., Linares, C., Díaz, J., 2012. Influence of local factors in the relationship between mortality and heat waves: Castile-La Mancha (1975–2003). *Sci. Total Environ.* 414, 73–80. <https://doi.org/10.1016/j.scitotenv.2011.10.009>.
- National Disaster Management Authority (NDMA) n.d. Heat Wave Available online: <https://www.ndma.gov.in/Natural-Hazards/Heat-Wave> (accessed on Oct 17, 2022).
- Nori-Sarma, A., Anderson, G.B., Rajiva, A., ShahAzhar, G., Gupta, P., Pednekar, M.S., Son, J.-Y., Peng, R.D., Bell, M.L., 2019. The impact of heat waves on mortality in Northwest India. *Environ. Res.* 176, 108546.
- Padhi, B.K., Padhy, P.K., 2008. Domestic fuels, indoor air pollution, and children's health: the case of rural India. *Ann. N. Y. Acad. Sci.* 1140, 209–217.
- Parks, R.M., Bennett, J.E., Tamura-Wicks, H., Kontis, V., Toumi, R., Danaei, G., Ezzati, M., 2020. Anomalously warm temperatures are associated with increased injury deaths. *Nat. Med.* 26, 65–70. <https://doi.org/10.1038/s41591-019-0721-y>.
- Rathi, S.K., Sodani, P.R., 2021. Summer temperature and all-cause mortality from 2006 to 2015 for Hyderabad, India. *Afr. Health Sci.* 21, 1474–1481.
- Rey, G., Fouillet, A., Bessemoulin, P., Frayssinet, P., Dufour, A., Jougla, E., Hémon, D., 2009. Heat exposure and socio-economic vulnerability as synergistic factors in heat-wave-related mortality. *Eur. J. Epidemiol.* 24, 495–502.
- Robinson, P.J., 2001. On the definition of a heat wave. *J. Appl. Meteorol. Climatol.* 40, 762–775.
- Sharma, A., Andhikaputra, G., Wang, Y.-C., 2022. Heatwaves in South Asia: characterization, consequences on human health, and adaptation strategies. *Atmosphere* 13.
- Sharma, A., Deng, L., Wang, Y.-C., 2023. Estimation of effects of extreme temperature on the risk of hospitalisation in Taiwan. *J. Epidemiol. Community Health* 77, 375–383.
- Sheridan, S.C., Lin, S., 2014. Assessing variability in the impacts of heat on health outcomes in New York City over time, season, and heat-wave duration. *EcoHealth* 11, 512–525. <https://doi.org/10.1007/s10393-014-0970-7>.
- Singh, N., Mhawish, A., Ghosh, S., Banerjee, T., Mall, R.K., 2019. Attributing mortality from temperature extremes: a time series analysis in Varanasi, India. *Sci. Total Environ.* 665, 453–464.
- Sixth Assessment Report of Intergovernmental Panel on Climate Change (IPCC) Climate Change 2021, The Physical Science Basis, 2021. Vol. IPCC AR6 W.
- Tong, S., FitzGerald, G., Wang, X.-Y., Aitken, P., Tippet, V., Chen, D., Wang, X., Guo, Y., 2015. Exploration of the health risk-based definition for heatwave: a multi-city study. *Environ. Res.* 142, 696–702.
- Urban, A., DiNapoli, C., Cloke, H.L., Kyselý, J., Pappenberger, F., Sera, F., Schneider, R., Vicedo-Cabrera, A.M., Acquaotta, F., Ragettli, M.S., 2021. Evaluation of the ERA5 reanalysis-based universal thermal climate index on mortality data in Europe. *Environ. Res.* 198, 111227.
- Wang, D., Lau, K.K.-L., Ren, C., Goggins, W.B.I.I.I., Shi, Y., Ho, H.C., Lee, T.-C., Lee, L.-S., Woo, J., Ng, E., 2019. The impact of extremely hot weather events on all-cause mortality in a highly urbanized and densely populated subtropical city: a 10-year time-series study (2006–2015). *Sci. Total Environ.* 690, 923–931. <https://doi.org/10.1016/j.scitotenv.2019.07.039>.
- Wei, Y., Tiwari, A.S., Li, L., Solanki, B., Sarkar, J., Mavalankar, D., Schwartz, J., 2021. Assessing mortality risk attributable to high ambient temperatures in Ahmedabad, 1987 to 2017. *Environ. Res.* 198, 111232.